EMP Firmware Architecture

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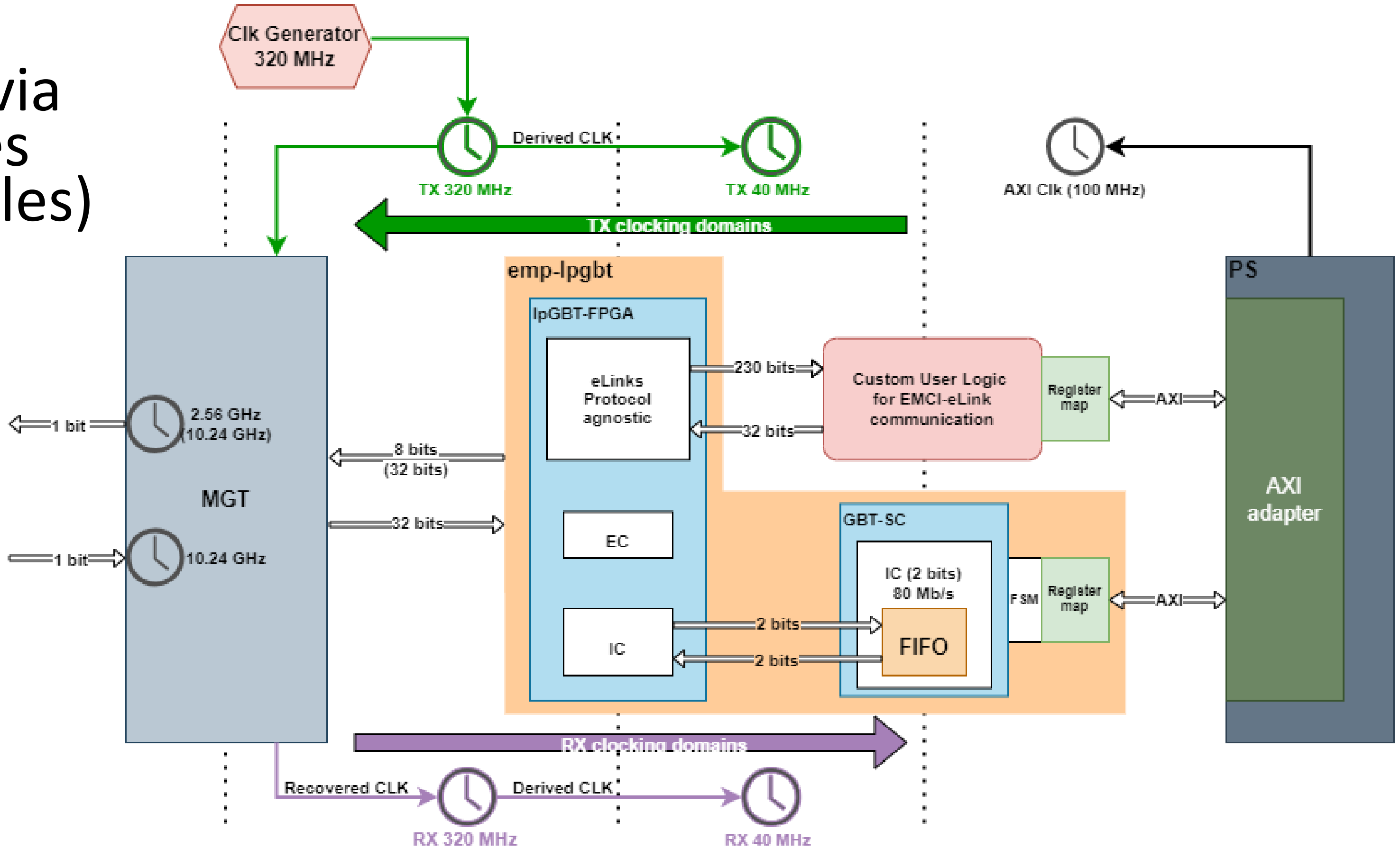
for the EMP Team

Overview

- Firmware architecture of the reference project
- IP cores:
  - emp-lpgbt
  - MGT
  - AirHDL register maps
  - Zynq Ultrascale+ MPSoC
  - Example for custom logic
- Current status
- Summary
- Pointers

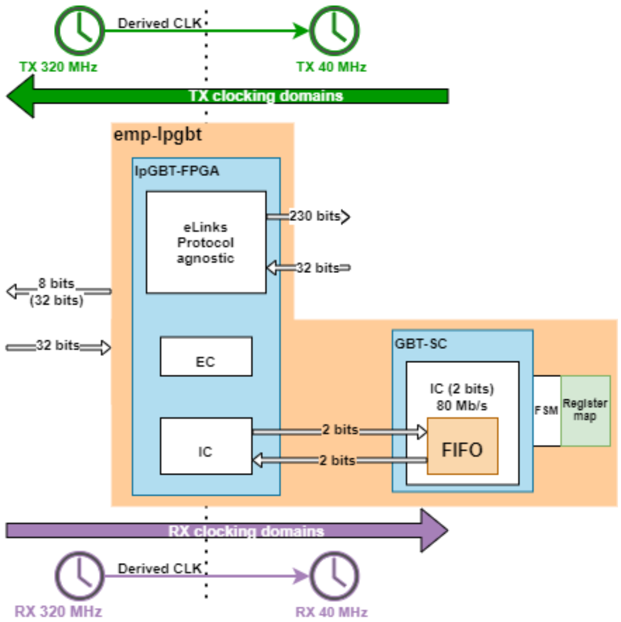
Firmware architecture of the reference project

- EMP firmware covers the communication path from PS (via register maps) to the MGT lanes (connected to the Firefly modules)
- Consists of several IPs:
  - emp-lpgbt (reference)
  - MGT
  - AirHDL register maps
  - Zynq Ultrascale+ MPSoC
  - Example for custom logic
- It is the recommended starting point for user related specific development



IP cores: emp-lpgbt

- The emp-lpgbt core serves as a <u>reference IP</u>
- It wraps the GBT-SC and lpGBT-FPGA IP to provide system specific functionality:
  - o Implements FSM for IC communication
  - o Implements AXI register map for read/write operations from PS



<u>lpGBT-FPGA</u>

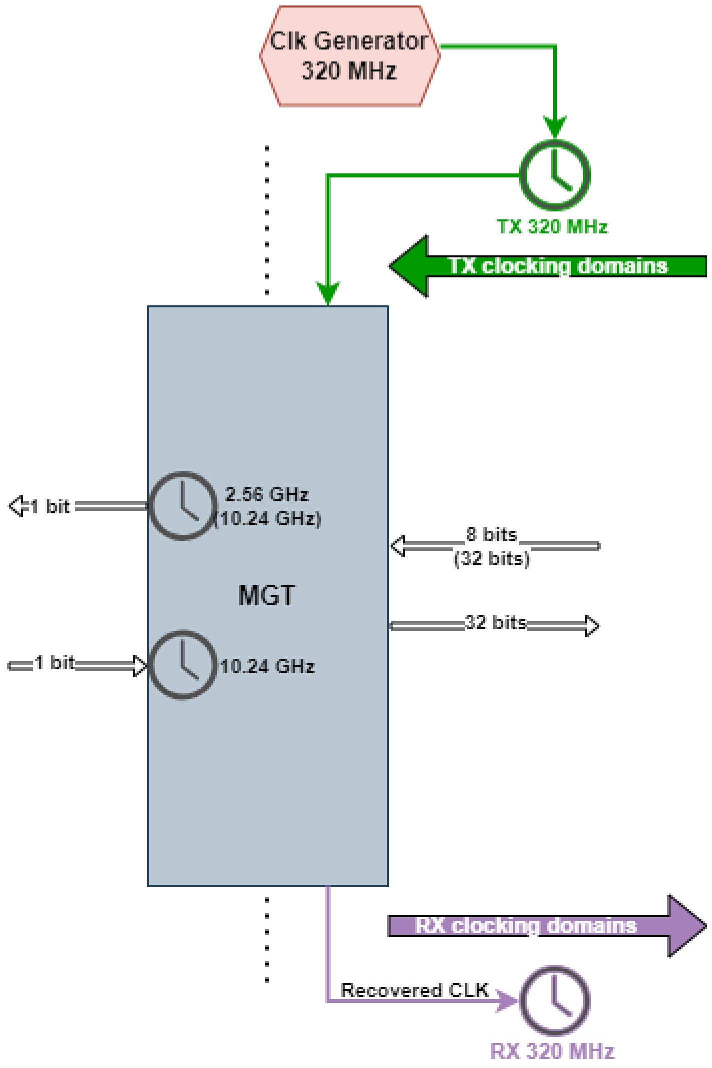
- Includes downlink and uplink data path to interface the lpGBT through its high-speed links
  - o Provides basic and manipulable signals for eLink communication
  - o IC and EC channel are still encoded in the frame
- Configured for uplink rate 10.24 Gbps and FEC5
- Supported by GBT team

<u>GBT-SC</u>

- Decodes the IC and EC channel frames
- Provides signals for reading/writing the lpGBT register (IC channel communication)
- FIFO storage based on RAM
- Supported by the GBT team

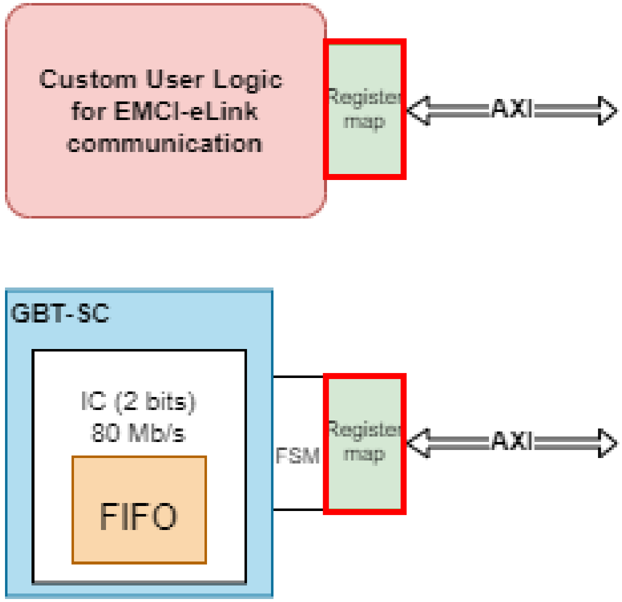
IP cores: MGT

- The Multi Gigabit Transceiver IP (for GTH) is configured to interface with 12 Firefly channels
  - Uplink : 10.24 Gbps
  - Downlink 10.24 Gbps (transmission replicated 4 times for emulating 2.56 Gbps)
- Based on KCU105 example of the GBT Team
- Clocking
  - TX path is clocked on the 320 MHz of the clock generator
  - RX path is clocked on the recovered clock from the GTH receiver
- Provides access to 12 Firefly channels



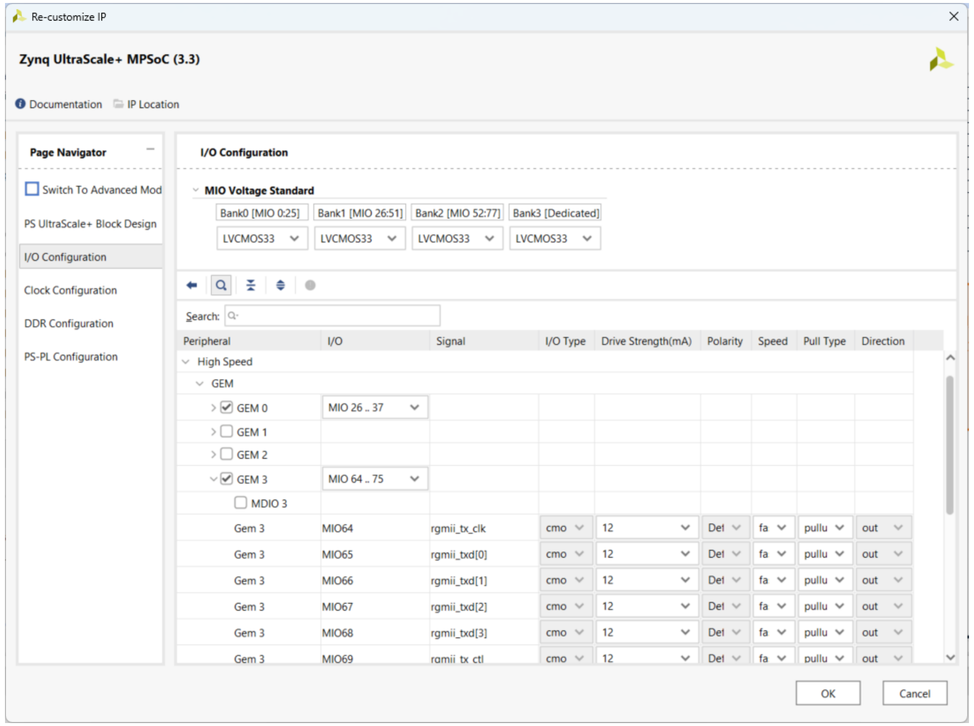
IP cores: AirHDL register maps

- airHDL is a tool to create register maps with AXI (https://airhdl.com)
- Generates automatically VHDL code which can be easily integrated in the design
- Register maps are used to provide an interface between firmware and PS
- Register can be read/written from PS to access/provide data to firmware



IP cores: Zynq Ultrascale+ MPSoC

- Vivado offers the possibility to pre-configure the PS
- Necessary for specific EMP functionality:
  - PS-PL communication (AXI interface)
  - Interrupts
  - MIO definitions (Ethernet, UART, SD, USB2, …)
  - GTR definitions (USB3)
- The definitions are exported via a hardware design (.xsa file) and used for the configuration of the boot files (generation of the device tree)

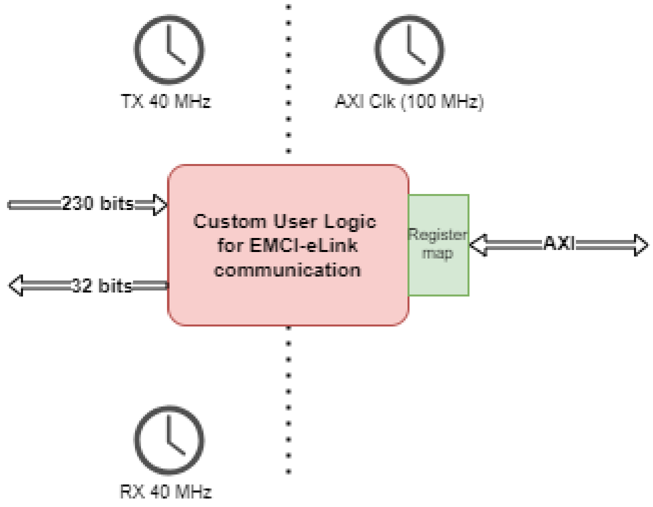


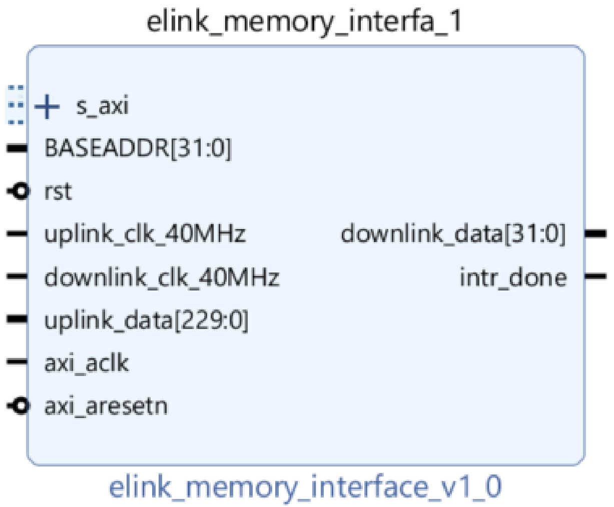
IP cores: Example for custom logic

- Many user applications will use lpGBT with the need of processing eLink data
- The eLink memory interface is a custom IP core to show an example on how to make use of eLinks (uplink/downlink)
- It interfaces the PS using a register map

➢ The IP should not simply be multiplied as the primitives used would exceed the resource utilization

➢ Users must develop their own logic





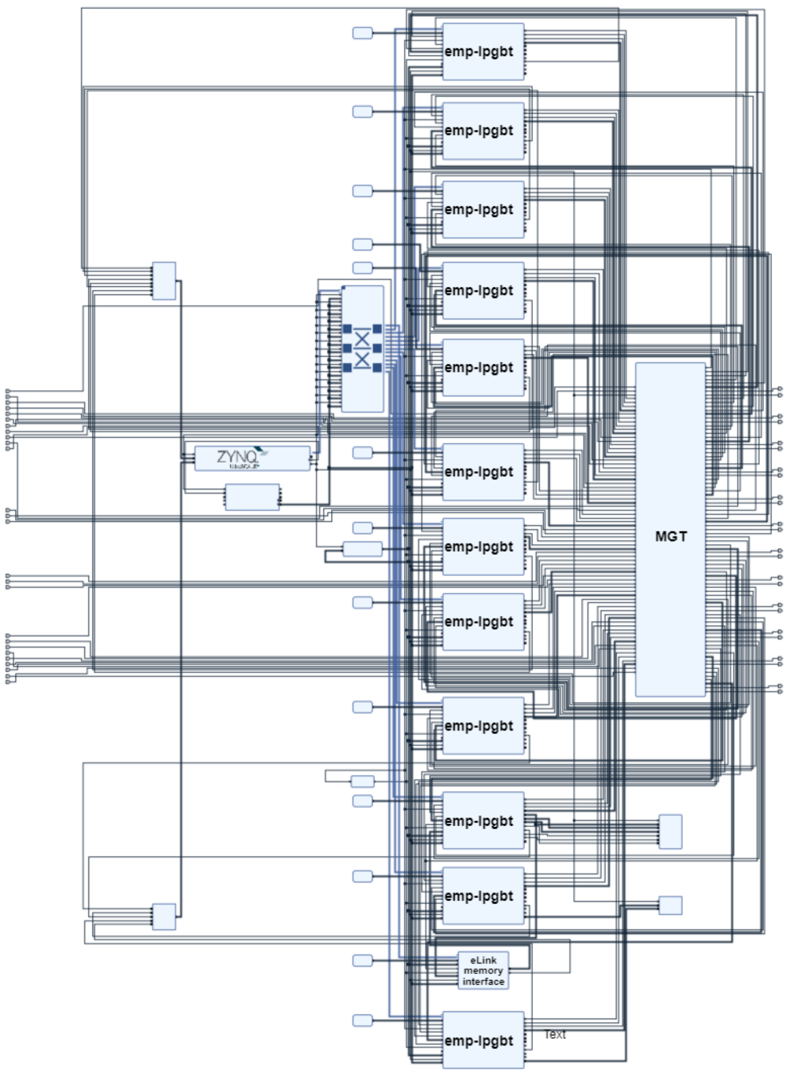
Current status

- Firmware tested on EMP baseboard as well as on Trenz baseboard
- HW description for both boards available on GitLab
- Tagged firmware version available for users on GitLab

emp_fw_v0.11
fa5bcc7a · Major clocking update. 40 MHz uplink clock for · 3 weeks ago

- Resource utilization for XCZU4EG-1FBVB900E

Resource	Utilization	Available	Utilization %
LUT	21164	87840	24.09
LUTRAM	540	57600	0.94
FF	38476	175680	21.90
BRAM	72.50	128	56.64
GT	12	16	75.00
BUFG	42	352	11.93
MMCM	2	4	50.00



Summary

- Current stable build available for users on GitLab
- Can be used as starting point for project specific development
- Users can:
  - Verify EMCI communication (read/write registers, read internal ADC, …)
  - Common IP blocks (e.g. emp-lpgbt) deliver useful functionality for eLink integration
  - The „eLink memory interface“ IP serves as an example how to use eLinks within the proposed firmware architecture
  - AXI configuration for IC communication available for PS and PL

- Ongoing work on firmware:
  - Optimization
  - Adding Features
  - Migrating to newer Vivado version

Pointers

- Firmware and IP cores
  - emp-firmware
  - GBT-SC
  - lpGBT-FPGA
- Register map creation
  - airHDL
- Guides (SPR of the EMP – „Firmware“ by Daniel Blasco Serrano):
  - Assembly procedure